



PROJECT IN MATHEMATICS

On Permutation Groups

by

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Abstract

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Chapter 1

Introduction

The motivation behind the research whose the results can be found largely within this work, was the desire to find an answer to a deceptively simple question: “What is the process by which methods for solving problems are found?”

The question arose within me in relation to a long-time fascination with the problem-solving process employed when finding methods for solving *twisty puzzles* – such as the *Rubik’s cube*. This focus led me to study *permutation groups*, and initially formulate my thoughts and findings in terms thereof.

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Chapter 2

Preliminaries

2.1 Groups

Definition 1 (Group). In mathematics, a *group* G denotes a set of elements $\{a, b, c, \dots\}$ and an accompanying binary operator $*$, together satisfying the following conditions:

1. *Closure*: For every ordered pair of elements in the group, the binary product thereof (as defined by the binary operator) is also an element of that group; $\forall a, b \in G \ (a * b \in G)$.
2. *Associativity*: The order wherein ordered pairs of elements are evaluated has no consequence on the resulting product; $\forall a, b, c \in G \ ((a * b) * c = a * (b * c))$.
3. *Existence of Identity*: There exists in the group an *identity* element, the omission of which from any expression containing any other element or elements has no consequence on the evaluation of that expression; $\exists e \in G \ (\forall a \in G \ (e * a = a * e = a))$.
4. *Existence of Inverse*: Each element in the group has its own *inverse*, who's product therewith evaluates to the identity; $\forall a \in G \ (\exists b \in G \ (a * b = e))$, where e is the identity.

Definition 2 (Commutativity). Let G be a group. Then it is *commutative* (or *Abelian*) if $\forall a, b \in G \ (a * b = b * a)$.

The integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ together with *addition* constitutes an Abelian group with 0 acting as its identity (since $\forall a \in \mathbb{Z} \ (0 + a = a + 0 = a)$). The negative numbers exist as the inverses of the positives, and vice versa. We know addition is associative, and since $\mathbb{Z}$ is infinite in each direction, we will never find a pair of elements therein whose sum is not also an element therein. This group can also be expressed as additively *generated* by the element 1, since 1 and its inverse -1 together with $+$ can express the whole group. This generative definition is denoted $G = (\langle\{1\}\rangle, +) = (\mathbb{Z}, +)$. The group generated by 2 – to give another example – is the group of all *even* integers.

Definition 3 (Cyclicity). A group which can be generated by a single element is called a *cyclic* group.

When the operator is given implicitly, a cyclic group with the generator g can be written as $\langle g \rangle$.

Definition 4 (Subgroup). Let G be a group under a binary operator $*$. Then $H \subset G$ is a *subgroup* of G if H is also a group under $*$.

2.2 Permutations

Definition 5 (Permutation). A *permutation* π on a set A is a bijective map $\pi : A \rightarrow A$.

We write π^n for a permutation π to denote $\pi \circ \dots \circ \pi$ where π is composed with itself $n \in \mathbb{N}$ times. π^{-n} denotes the composition $\pi^{-1} \circ \dots \circ \pi^{-1}$ where the inverse π^{-1} is composed with itself $n \in \mathbb{N}$ times. The permutation π^0 always equals the identity.

Definition 6 (Permutation Group). A *permutation group* on a set A is a group of permutations on A under composition.

Definition 7 (Support). The *support* $\underline{\pi}$ of a permutation π on A is the set $\{a \in A \mid \pi(a) \neq a\}$.

A permutation whose support is the empty set equals the identity. We say that a permutation π on A *fixes* an element $a \in A$ if $a \notin \underline{\pi}$.

Definition 8 (Cycle). A *cycle* is a permutation π such that $\forall a, b \in \underline{\pi} \ (\exists \phi \in \langle \pi \rangle \ (\phi(a) = b))$.

Since the support of the identity is the empty set, it follows that the identity also a cycle.

Permutations are commonly expressed using *cycle notation*, wherein each cycle is written as a collection of parenthesized series of elements. For example, a permutation π on $\{1, 2, 3, 4, 5\}$ defined by

$$\pi = \left\{ \begin{array}{l} 1 \mapsto 2 \\ 2 \mapsto 3 \\ 3 \mapsto 1 \\ 4 \mapsto 5 \\ 5 \mapsto 4 \end{array} \right\}$$

is expressed by its cycles like

$$\begin{aligned} \pi &= (1, 2, 3)(4, 5) \\ &= (2, 3, 1)(5, 4) \\ &= (4, 5)(3, 1, 2) \\ &\vdots \end{aligned}$$

where each element is mapped to the succeeding element save for the last one, which is mapped to the first. When writing about permutations on elements 0 through 9, the commas inbetween the numbers can be omitted like so: $(4, 6, 8, 2) = (4682)$.

Definition 9 (Set Form of a Permutation). The *set form* $\tilde{\pi}$ of a permutation π on A is the set of cycles on A with pairwise disjoint supports where $\forall a \in \underline{\pi} \ (\exists \phi \in \tilde{\pi} \ (\phi(a) = \pi(a)))$ and $\bigcup_{\phi \in \tilde{\pi}} \phi = \underline{\pi}$.

Example 1. Let $\pi = (12)(5349)(687)$. Then $\tilde{\pi} = \{(12), (5349), (687)\}$.

Definition 10 (Permutation Form of a Set). The *permutation form* $P(S)$ of a set of cycles S on A (with pairwise disjoint supports) is the permutation on A whose set form is S .

When writing about the permutation form of a subset (or any other set relation) of the set form of a permutation, we use a shorthand: we write $\phi \subset \pi$ – for example – to mean $\phi = P(S)$ where $S \subset \tilde{\pi}$.

Example 2. Let $S = \{(12), (34), (56), (789)\}$. Then $P(S) = (12)(34)(56)(789)$.

Definition 11 (Subpermutation). The *subpermutation* $\overset{S}{\pi}$ of a permutation π for a set $S \subset \underline{\pi}$ is the permutation $\overset{S}{\pi} \subset \pi$ where $S \subseteq \underline{\pi}$.

A subpermutation which is a cycle is, conversely, called a *subcycle* – take for instance $\overset{\{a\}}{\pi} \in \pi$, where $a \in \underline{\pi}$.

Example 3. Let $\pi = (12)(34)(56)(789)$ and $\phi = (34)(789)$. Then $\phi \subset \pi$ and $\overset{\{3,7\}}{\pi} = \overset{\{2,3,7\}}{\pi} \dots$.

Definition 12 (Orbit). The *orbit* of an element $a \in A$ in a permutation π on A is the set $\overset{\{a\}}{\underline{\pi}}$.

Example 4. Let $\pi = (253)(798)$. Then the orbit of 8 in π is the set $\{7, 8, 9\}$.

Definition 13 (Transposition). A *transposition* is a cycle π with $|\underline{\pi}| = 2$.

Example 5. Let $\pi = (19)(37)(58)$. Then each cycle in $\tilde{\pi}$ is a transposition, since $\forall \phi \in \tilde{\pi} (\phi)$.

Definition 14 (Order of a permutation). The *order* $|\pi|$ of a permutation π on A with $2 \leq |\underline{\pi}| < \infty$ is the smallest number $n \in \mathbb{N}$ (where $\mathbb{N} = \{1, 2, 3, \dots\}$) such that $\pi^n = \varepsilon$, where $\varepsilon : A \rightarrow A$ is the identity $\varepsilon(x) = x$.

Lemma 1. Let ϕ be a cycle. Then the order of ϕ is $|\underline{\phi}|$.

Proof. Let $a \in \underline{\phi}$. Then since $|\underline{\pi}| \geq 2$, $\phi(a) = b$ where $b \in \underline{\phi}$ and $a \neq b$. If $\phi(b) = a$, then $\phi = \{a, b\}$ and the order is 2, since $\phi(\phi(x)) = \phi^2(x) = x$ for $x \in \{a, b\}$. Now, let $\alpha = (a_1, \dots, a_k)$ have the order n and $\beta = (b_1, \dots, b_k, b_{k+1})$ have the order m . Then $\alpha^{n-1}(a_1) = a_k$ and $\beta^{n-1}(b_1) = b_k$. So $\beta^n(b_1) = b_{k+1}$ and $\beta^{n+1}(b_1) = b_1$, wherefore $m = n + 1$. By induction, then, follows the lemma herefrom. $\square$

Theorem 1. Let π be a permutation where $\underline{\pi}$ is finite. Then the order of π is the least common multiple of the order of its cycles $\phi \in \tilde{\pi}$.

Proof. Let $\phi \in \tilde{\pi}$ have the order n . Then $\phi^{kn} = \phi^n = \varepsilon$ for all $k \in \mathbb{N}$ (where ε is the identity), since $\phi^{kn} = \overbrace{\phi^n \circ \dots \circ \phi^n}^k = \overbrace{\varepsilon \circ \dots \circ \varepsilon}^k = \varepsilon$. Now, let $S = \{|\phi| \mid \phi \in \tilde{\pi}\}$. Then $\forall \phi \in \tilde{\pi} (\phi^{\text{lcm}(S)} = e)$, since $\exists k \in \mathbb{N} (k \cdot |\phi| = \text{lcm}(S))$ for all $\phi \in \tilde{\pi}$. Because every cycle in $\pi^{\text{lcm}(S)}$ is the identity, it follows that $\pi^{\text{lcm}(S)}$ is the identity also. The fact that $\text{lcm}(S)$ is the *smallest* number with this property follows from the definition of the *least common multiple*. $\square$

Example 6.

$$|(123)(45)| = 6$$

$$|(123)(456)(789)| = 3$$

$$|(012345)(6789)| = 12$$

Chapter 3

The Conjugates and Commutators of Permutations

Herein, all lowercase geek letters are implied to be elements in a permutation group G on A . Elements of A are represented by lowercase latin characters. Further, the operator $\circ$ will henceforth be implicit, and thus omitted from expressions. As an example, take $\pi \circ \phi = \pi\phi$.

Definition 15 (Conjugate). The *conjugate* $\searrow_{\beta}^{\alpha}$ of β by α is the composition $\alpha^{-1}\beta\alpha$.

Definition 16 (Commutator). The *commutator* $\times_{\beta}^{\alpha}$ of α and β is the composition $\beta^{-1}\alpha^{-1}\beta\alpha$.

These two constructions – though they can seem arbitrary at first – have interesting and useful properties when working within groups – specifically when the supports of their two permutations share elements.

In the trivial case where they share no element, the permutations commute, so the conjugate of α and β is β , and the commutator thereof is the identity; $\alpha \cap \beta = \emptyset$ implies that $\alpha^{-1}\beta\alpha = \beta$ and $\beta^{-1}\alpha^{-1}\beta\alpha = e$.

Lemma 2. Let $\underline{\alpha} \cap \underline{\beta} = \{c\}$. Then $\searrow_{\beta}^{\alpha} = (\alpha^{-1}(c), \beta(c), \dots, \beta^{-1}(c))$.

Proof. For each $b \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\}$, $\alpha(b) \notin \underline{\beta}$. Since $\beta\alpha(b) = \alpha(b)$, the conjugate can be rewritten for b : $(\alpha^{-1}\beta\alpha)(b) = (\alpha^{-1}\alpha)(b) = b$. In the case of $\alpha^{-1}(c)$, $(\alpha^{-1}\beta\alpha\alpha^{-1})(c) = (\alpha^{-1}\beta)(c) = \beta(c)$, since $\beta(c) \notin \underline{\alpha} \implies (\alpha^{-1}\beta)(c) = \beta(c)$. For each $d \in \underline{\beta} \setminus \{\beta^{-1}(c), c\}$, $d, \beta(d) \notin \underline{\alpha}$, so $(\alpha^{-1}\beta\alpha)(d) = \beta(d)$. Lastly, $\beta^{-1}(c)$ maps to $(\alpha^{-1}\beta\alpha\beta^{-1})(c) = (\alpha^{-1}\beta\beta^{-1})(c) = \alpha^{-1}(c)$. This gives the function

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \underline{\beta} \setminus \{\beta^{-1}(c), c\} \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \quad (3.1)$$

whose subcycle of c is written as $(\alpha^{-1}(c), \beta(c), \dots, \beta^{-1}(c))$. □

Corollary 1. If β is a transposition, then so is $\searrow_{\beta}^{\alpha}$.

Proof. If β is a transposition, then $\beta = (c, \beta(c))$ and $\beta(c) = \beta^{-1}(c)$. Using (3.1) we get

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \beta(x) & : x \in \emptyset \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \end{cases} \quad (3.2)$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ \beta(c) & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta(c) \end{cases} \quad (3.3)$$

$$= (\alpha^{-1}(c), \beta(c)). \quad (3.4)$$

□

Lemma 3. Let $\underline{\alpha} \cap \underline{\beta} = \{c\}$. Then $\times_{\beta}^{\alpha} = (c, \beta^{-1}(c), \alpha^{-1}(c))$.

Proof. Using (3.1) from Lemma 2, we get

$$\times_{\beta}^{\alpha}(x) = (\beta^{-1} \searrow_{\beta}^{\alpha})(x) = \begin{cases} (\beta^{-1})(x) & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c)\} \\ (\beta^{-1}\beta)(c) & : x = \alpha^{-1}(c) \\ (\beta^{-1}\beta)(x) & : x \in \underline{\beta} \setminus \{\beta^{-1}(c), c\} \\ (\beta^{-1}\alpha^{-1})(c) & : x = \bar{\beta}^{-1}(c) \end{cases} \quad (3.5)$$

$$= \begin{cases} x & : x \in \underline{\alpha} \setminus \{\alpha^{-1}(c), c\} \\ \beta^{-1}(x) & : x = c \\ c & : x = \alpha^{-1}(c) \\ x & : x \in \underline{\beta} \setminus \{\beta^{-1}(c), c\} \\ \alpha^{-1}(c) & : x = \bar{\beta}^{-1}(c) \end{cases} \quad (3.6)$$

$$= \begin{cases} x & : x \in (\underline{\alpha} \cup \underline{\beta}) \setminus \{\alpha^{-1}(c), \beta^{-1}(c), c\} \\ c & : x = \alpha^{-1}(c) \\ \alpha^{-1}(c) & : x = \beta^{-1}(c) \\ \beta^{-1}(c) & : x = c \end{cases} \quad (3.7)$$

$$= (c, \beta^{-1}(c), \alpha^{-1}(c)). \quad (3.8)$$

□

Lemma 4. Let $\underline{\alpha} \cap \underline{\beta} = \{c_1, \dots, c_n\}$, $\alpha \neq \beta$ for $n \geq 2$ where $\alpha^{k-1}(c_1) = \beta^{k-1}(c_1) = c_k$ for $k \in \{1, \dots, n\}$. Then $\searrow_{\beta}^{\alpha} = (\alpha^{-1}(c_1), c_1, \dots, c_{n-1}, \beta(c_n), \dots, \beta^{-1}(c_1))$.

Proof. Let $a \in \{c_n, \alpha(c_n), \dots, \alpha^{-2}(c_1)\}$. Then $\alpha(a) \notin \underline{\beta}$, so $(\alpha^{-1}\beta\alpha)(a) = (\alpha^{-1}\alpha)(a) = a$. Conversely, each $b \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\}$ maps to $(\alpha^{-1}\beta\alpha)(b) = \beta(b)$, since $b, \beta(b) \notin \underline{\alpha}$. As for the elements $\{c_1, \dots, c_{n-2}\}$, none thereof diverge from the common path $\alpha^{k-1}(c_1) =$

$\beta^{k-1}(c_1) = c_k$ for $k \in \{1, \dots, n\}$, wherefore $c_j \xrightarrow{\alpha} c_{j+1} \xrightarrow{\beta} c_{j+2} \xrightarrow{\alpha^{-1}} c_{j+1}$ in $\searrow_{\beta}^{\alpha}$ for each $j \in \{1, \dots, n-2\}$. Lastly, $\alpha^{-1}(c_1)$ maps to $(\alpha^{-1}\beta\alpha\alpha^{-1})(c_1) = (\alpha^{-1}\beta)(c_1) = \alpha^{-1}(c_2) = c_1$, $\beta^{-1}(c_1)$ maps to $(\alpha^{-1}\beta\alpha\beta^{-1})(c_1) = (\alpha^{-1}\beta\beta^{-1})(c_1) = \alpha^{-1}(c_1)$ and c_{n-1} maps to $(\alpha^{-1}\beta\alpha)(c_{n-1}) = (\alpha^{-1}\beta)(c_n) = \beta(c_n)$. This gives the function

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} x & : x \in \{c_n, \alpha(c_n), \dots, \alpha^{-2}(c_1)\} \\ c_1 & : x = \alpha^{-1}(c_1) \\ \beta(x) & : x \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\} \cup \{c_1, \dots, c_{n-2}\} \\ \beta(c_n) & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \quad (3.9)$$

$$\implies \searrow_{\beta}^{\alpha} = \left(\alpha^{-1}(c_1), c_1, \dots, c_{n-1}, \beta(c_n), \dots, \beta^{-1}(c_1) \right). \quad (3.10)$$

□

Lemma 5. Let α and β be permutations with the same properties as those in Lemma 4. Then $\searrow_{\beta}^{\alpha} = (\alpha^{-1}(c_1), \beta^{-1}(c_1))(c_{n-1}, c_n)$.

Proof. From (3.9) from Lemma 4 we get

$$\searrow_{\beta}^{\alpha}(x) = \begin{cases} \beta^{-1}(x) & : x \in \{c_n, \alpha(c_n), \dots, \alpha^{-2}(c_1)\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ (\beta^{-1}\beta)(x) & : x \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\} \cup \{c_1, \dots, c_{n-2}\} \\ (\beta^{-1}\beta)(c_n) & : x = c_{n-1} \\ (\beta^{-1}\alpha^{-1})(c_1) & : x = \beta^{-1}(c_1) \end{cases} \quad (3.11)$$

$$= \begin{cases} x & : x \in \{\alpha(c_n), \dots, \alpha^{-2}(c_1)\} \\ \beta^{-1}(c_n) & : x = c_n \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ x & : x \in \{\beta(c_n), \dots, \beta^{-2}(c_1)\} \cup \{c_1, \dots, c_{n-2}\} \\ c_n & : x = c_{n-1} \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \end{cases} \quad (3.12)$$

$$= \begin{cases} x & : x \in \{c_{n-1}, c_n, \alpha^{-1}(c_1), \beta^{-1}(c_1)\} \\ \beta^{-1}(c_1) & : x = \alpha^{-1}(c_1) \\ \alpha^{-1}(c_1) & : x = \beta^{-1}(c_1) \\ c_n & : x = c_{n-1} \\ c_{n-1} & : x = c_n \end{cases} \quad (3.13)$$

$$= \left(\alpha^{-1}(c_1), \beta^{-1}(c_1) \right) (c_{n-1}, c_n). \quad (3.14)$$

□

Lemma 6. Let α and β be permutations such that $\{\alpha(b) \mid b \in \underline{\beta}\} \cap \underline{\beta} = \emptyset$ and .

DEFINE PERMUTATION SHAPE.

Lemma 7. Let $\emptyset \neq \underline{\alpha} \cap \underline{\beta} \neq \underline{\alpha} \cup \underline{\beta}$ and $\forall \beta' \in \beta \left(\underline{\beta'} \subseteq \underline{\alpha} \vee \underline{\beta'} \cap \underline{\alpha} = \emptyset \right)$. Now, let γ be a permutation $\gamma \subset \beta$ such that $\underline{\beta \setminus \gamma} \cap \underline{\alpha} = \emptyset$ and $\underline{\gamma} \subseteq \underline{\alpha}$. Then $\overrightarrow{\times}_{\beta}^{\alpha} = \overrightarrow{\times}_{\gamma}^{\alpha}$? Then the shape of γ^{-1} is applied on $\underline{\gamma}$ and γ is applied on $\left\{ \alpha(g) \mid g \in \underline{\gamma} \right\}$ by the commutator $\overrightarrow{\times}_{\beta}^{\alpha}$.

Proof. Let $\beta = \beta_1, \dots, \beta_n \beta'_1, \dots, \beta'_m$ and $\beta' = \beta'_1, \dots, \beta'_m$. Then

$$\overrightarrow{\times}_{\beta}^{\alpha} = \begin{cases} x & : x \in \underline{\beta} \setminus \underline{\beta'} \\ \beta'(x) & : x \end{cases}.$$

□

Chapter 4

The 2×3 Game

The “ 2×3 game” is a simple puzzle with six tiles in a grid of numbers and a set of allowed moves which permute them. The goal of the game is to sort the grid into a “solved” state from some scrambled state.

$$\begin{bmatrix} 5 & 4 & 2 \\ 3 & 6 & 1 \end{bmatrix} \xrightarrow{\text{Move}_1} \cdots \xrightarrow{\text{Move}_n} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

Formally, the game can and will herein henceforth be modelled as follows. Let $G := (\langle \mathfrak{M} \rangle, \circ)$ be a group compositionally generated by the set of moves $\mathfrak{M} := \{(123), (456), (14)\}$, named $U = (123)$ for *upper row*, $L = (456)$ for *lower row* and $S = (14)$ for *side flip*. A *problem* is a permutation $p \in G$, and a *solution* thereto is a vector $\mathbf{s} \in \mathfrak{M}^*$ such that

$$\bigcirc_{i=n}^1 \mathbf{s}_i = \mathbf{s}_n \circ \mathbf{s}_{n-1} \circ \cdots \circ \mathbf{s}_1 = p$$

where n denotes the length of $\mathbf{s}$. Note that $\mathbf{s}$ may include the implicitly given inverses of the group generators. Hereafter, $\bigcirc_{i=n}^1 \mathbf{s}_i$ will be written as $\acute{\mathbf{s}}$.

This group equals the symmetric group S_6 . **MAKE THIS A THEOREM/LEMMA. AND MAKE THE FOLLOWING ITS PROOF:** This fact can be proven in the following way. Each transposition between the upper and lower row can be found with the nested conjugate

$$\begin{aligned} T_1(x, y) &= \left(L^{-y+3+1} \right)^{-1} \left(\left(U^{-x+1} \right)^{-1} (S) \left(U^{-x+1} \right) \right) \left(L^{-y+3+1} \right) \\ &= L^{y-4} U^{x-1} S U^{-x+1} L^{-y+4} \\ &= L^{y-1} U^{x-1} S U^{-x+1} L^{-y+1}, \end{aligned}$$

where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$. As an example, let $p = (26)$. Then

$$\begin{aligned}
T_1(2, 6) &= L^{6-1}U^{2-1}SU^{-2+1}L^{-6+1} \\
&= L^5U^1SU^{-1}L^{-5} \\
&= (456)^5(123)^1(14)(123)^{-1}(456)^{-5} \\
&= (465)(123)(14)(132)(456) \\
&= (465)(24)(456) \\
&= (2, 6).
\end{aligned}$$

This gives – after some reductions – the solution $\mathbf{s} = (L, U^{-1}, S, U, L^{-1})$ visualized below.

$$\begin{array}{ccccc}
\boxed{\begin{matrix} 1 & 6 & 3 \\ 4 & 5 & 2 \end{matrix}} & \xrightarrow{L} & \boxed{\begin{matrix} 1 & 6 & 3 \\ 2 & 4 & 5 \end{matrix}} & \xrightarrow{U^{-1}} & \boxed{\begin{matrix} 6 & 3 & 1 \\ 2 & 4 & 5 \end{matrix}} \\
\xrightarrow{S} \boxed{\begin{matrix} 2 & 3 & 1 \\ 6 & 4 & 5 \end{matrix}} & \xrightarrow{U} & \boxed{\begin{matrix} 1 & 2 & 3 \\ 6 & 4 & 5 \end{matrix}} & \xrightarrow{L^{-1}} & \boxed{\begin{matrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{matrix}}
\end{array}$$

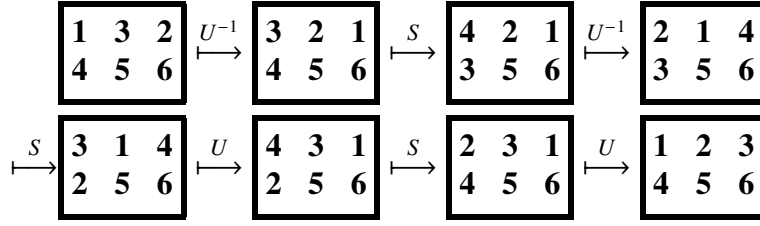
Each transposition between elements on the same row can be found with the nested conjugate

$$\begin{aligned}
T_2(x, y, R) &= \left((R^{-x+1})^{-1}(S)(R^{-x+1}) \right)^{-1} \left((R^{-y+1})^{-1}(S)(R^{-y+1}) \right) \left((R^{-x+1})^{-1}(S)(R^{-x+1}) \right) \\
&= (R^{x-1}SR^{-x+1})^{-1}(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{-(x+1)}S^{-1}R^{-(x-1)})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= (R^{x-1}SR^{-x+1})(R^{y-1}SR^{-y+1})(R^{x-1}SR^{-x+1}) \\
&= R^{x-1}SR^{-x+1}R^{y-1}SR^{-y+1}R^{x-1}SR^{-x+1} \\
&= R^{x-1}SR^{-x+y}SR^{x-y}SR^{-x+1}
\end{aligned}$$

where $x, y \in R$ and $R \in \mathfrak{M} \setminus \{S\}$. As an example, let $p = (23)$. Since 2 and 3 belong to the upper row, R is set to U :

$$\begin{aligned}
T_2(2, 3, U) &= U^{2-1}SU^{-2+3}SU^{2-3}SU^{-2+1} \\
&= U^1SU^1SU^{-1}SU^{-1} \\
&= (123)(14)(123)(14)(123)^{-1}(14)(123)^{-1} \\
&= (123)(14)(123)(14)(132)(14)(132) \\
&= (123)(14)(24)(14)(132) \\
&= (123)(12)(132) \\
&= (23).
\end{aligned}$$

This gives the solution $\mathbf{s} = (U^{-1}, S, U^{-1}, S, U, S, U)$ visualized below.



Since a transposition of any two elements herein can be composed using the generators in $\mathfrak{M}$ – in other words, since the group G includes every transposition of the elements 1 through 6 – it follows that the group must include every permutation on the six elements, and thereby equal S_6 .

Since G is the symmetric group S_6 , there must exist a map $T_1(x, y) = (x, y)$, where $x \in \{1, 2, 3\}$ and $y \in \{4, 5, 6\}$.

From Corrolary 1, we know that since $\zeta_1(x, y)$ is a transposition, then $\zeta_1 = \searrow_{\beta_1}^{\alpha_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$. Similarly, from $|\underline{\beta}_1| = 2$ follows that $\beta_1 = \searrow_{\beta_2}^{\alpha_2}$ where $\beta_2 = (c_1, c_2)$ and $\alpha_2(x) = c_2$. Now, from $\mathfrak{M}$ we can find permutations matching these properties

$$T_1(x, y) = \searrow_{\substack{U^{-x+1} \\ S}}^{L^{-y+3+1}}$$

$$\begin{aligned}
 T_2(x, y, R) &= \left((R^{-x+1})^{-1} (S) (R^{-x+1}) \right)^{-1} \left((R^{-y+1})^{-1} (S) (R^{-y+1}) \right) \left((R^{-x+1})^{-1} (S) (R^{-x+1}) \right) \\
 &= R^{x-1} S R^{-x+y} S R^{x-y} S R^{-x+1}
 \end{aligned}$$

Since G is the symmetric group S_6 , there must be a map $\zeta : \{1, \dots, 6\}^2 \rightarrow \mathcal{P}(\mathfrak{M}^{<\omega})$, where $\mathfrak{M}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathfrak{M}^n$ and $s \in \zeta(x, y) \implies \acute{s} = (x, y)$. In other words: there must exist a set of solutions to each transposition in S_6 .

Since G is the symmetric group S_6 , there must be an infinite¹ set of maps $\{1, \dots, 6\}^2 \rightarrow \mathfrak{M}^{<\omega}$, where $\mathfrak{M}^{<\omega} = \bigcup_{n \in \mathbb{N}} \mathfrak{M}^n$ and $s \in \zeta(x, y) \implies \acute{s} = (x, y)$ where ζ is such a map.

There is a set of solutions to T_1 , and α_1 and β_1 and α_2 and β_2 . the solution is the intersection thereof. At each level there is a unique map (for that level) to a set of solutions². It's recursive!

—
The lemmas and corollaries in chapter 3 – and indeed, any other observation, general or particular (about permutations or a specific game) – together construct a formal language.

Example: π where $|\underline{\pi}| = 2$ is a *pattern*, which can be substituted by another pattern: a conjugation $\searrow_{\beta_1}^{\alpha_1}$ where $\beta_1 = (x, c_1)$ and $\alpha_1(y) = c_1$ (as in the example above). The permutation $\beta_1 = (x, c_1)$ is, then, also a pattern to be substituted and so on until you have a permutation constructed from available moves ($\mathfrak{M}?$). You don't even need to evaluate them all if you know, for example, that ζ_1/T_1 exists.

All the lemmas and corollaries are in fact rewritings (with rewriting rules – in this case, formal math) ...

A *pattern* is a set p . If a set, vector, element (or other data) Ω matches the pattern p , then $\Omega \in p$. The set $\{p \mid \Omega \in p\}$ is the set of patterns to which some data Ω belongs.

Example

Let $a \gg b = [a_1, \dots, a_n, b_1, \dots, b_m]$, where $a = [a_1, \dots, a_n]$ and $b = [b_1, \dots, b_m]$.

$$\begin{aligned} \mathfrak{E} &:= \mathfrak{T} \cup \{e_1 \gg o \gg e_2 \mid o \in \{[“+”], [“-”]\} \wedge e_1, e_2 \in \mathfrak{E}\} \\ \mathfrak{T} &:= \mathfrak{F} \cup \{t_1 \gg o \gg t_2 \mid o \in \{[“\times”], [“\div”]\} \wedge t_1, t_2 \in \mathfrak{T}\} \\ \mathfrak{F} &:= \mathfrak{N} \cup \{[“(”] \gg e \gg [“)”] \mid e \in \mathfrak{E}\} \\ \mathfrak{N} &:= \mathcal{N} \cup \mathcal{N}^* \\ \mathcal{N} &:= \{“0”, \dots, “9”\} \end{aligned}$$

¹It is easy to see that this set is infinite, since one can append any solution with a permutation and its inverse to get a new solution. Take the solution $s_2 = (\alpha, \beta, \gamma)$ as an example, then $\acute{s}_1 = \acute{s}_2$ where $s_2 = (\alpha, \beta, \gamma, \alpha, \alpha^{-1})$.

²These are *patterns* with rewriting rules!

Chapter 5

Finding Methods

Definition 17 (Grimoire). A *grimoire* Φ of a semigroup G , a set of *spells* $\mathfrak{S}_\Phi \subseteq G$ and a set of *movements* $\mathfrak{M} \subseteq \mathfrak{S}_\Phi$ is a map in $\mathfrak{S}_\Phi \rightarrow \mathcal{P}(\mathfrak{S}_\Phi^{<\omega})$ such that $s_1 * \dots * s_n = s$ – where $*$ is the binary operator belonging to G – for each $[s_1, \dots, s_n] \in \Phi(s)$ and $s \in \mathfrak{S}_\Phi \setminus \mathfrak{M}$.

Definition 18 (Exploration Function). Let a map $\Delta : \boxed{\Phi} \times G^{<\omega} \rightarrow \boxed{\Phi}$ – where $\boxed{\Phi}$ is the set of all grimoires of a subgroup G – be defined for some grimoire Φ by $\forall s \in \mathfrak{S}_\Phi \setminus \{s_1 * \dots * s_n\} (\Phi'(s) = \Phi(s))$ and $\Phi'(s_1 * \dots * s_n) = \Phi(s_1 * \dots * s_n) \cup \{[s_1, \dots, s_n]\}$ **!!!MAKE MORE CLEAR WHAT IS INCLUDED IN THE "IF" STATEMENT ("AND"???)!!!** if $s_1 * \dots * s_n \in \mathfrak{S}_\Phi$ and $\Phi'(s_1 * \dots * s_n) = \{[s_1, \dots, s_n]\}$ if not – where $\Phi' = \Delta(\Phi, [s_1, \dots, s_n])$ and $s_1, \dots, s_n \in \mathfrak{S}_\Phi$. (Note that since $\mathfrak{S}_\Phi = \text{dom}(\Phi)$, then $\mathfrak{S}_{\Phi'} = \mathfrak{S}_\Phi \cup \{s_1 * \dots * s_n\}$ if $\Phi' = \Delta(\Phi, [s_1, \dots, s_n])$.) An *exploration function* $\mathcal{E}$ in G is a map in $\boxed{\Phi} \times \boxed{C}^{<\omega} \rightarrow \boxed{\Phi}$ defined by

$$\mathcal{E}(\Phi, [C_1, \dots, C_n]) := \bigcirc_{i=1}^n \Delta(\Phi, C_i(\mathfrak{S}_\Phi)),$$

!!!THE EXPLORATION IS NON-RECURSIVE!!! where $\boxed{C} \subset \mathcal{P}(G) \rightarrow G^{<\omega}$ and **!!!THE DEFINITION OF C IS SLOPPY!!!** $C(S) \in S^{<\omega}$ for each $C \in \boxed{C}$ and $S \in \mathcal{P}(G)$.

Definition 19 (Expansion Function). An *expansion function* λ for a grimoire Φ of a semigroup G is a map in $\boxed{\Phi} \times \mathfrak{S}_\Phi \times \boxed{\Pi} \rightarrow \mathfrak{M}^{<\omega}$ – where $\boxed{\Phi}$ is the set of all grimoires of G – defined by

$$\lambda(\Phi, s, \Pi) := \begin{cases} s & : \lambda_{\text{step}}(\Phi, s, \Pi) = s \\ \lambda(\Phi, \lambda_{\text{step}}(\Phi, s, \Pi), \Pi) & : \text{otherwise} \end{cases}$$

where $\lambda_{\text{step}} : \boxed{\Phi} \times \mathfrak{S}_\Phi \times \boxed{\Pi} \rightarrow \mathfrak{S}_\Phi^{<\omega}$ is defined by

$$\lambda_{\text{step}}(\Phi, s, \Pi) := \bigvee_{i=1}^{|s|} \left(\begin{cases} s_i & : s_i \in \mathfrak{M} \\ \Pi(\Phi(s_i)) & : \text{otherwise} \end{cases} \right),$$

where $\boxed{\Pi} \subset \mathcal{P}(G^{<\omega}) \rightarrow G^{<\omega}$ is a set of maps wherein $\Pi(S) \in S$ holds for each $\Pi \in \boxed{\Pi}$ and $S \in \mathcal{P}(G^{<\omega})$.

Proove that grimoire-exploration with $\mathcal{E}$ or Δ is a semigroup.

Är gruppen definierad av *likheten* mellan en mängd utforskningar/utökningar av trollformelboken?